Answers on
Booklet - 5

Taken from OCR Physics A past papers

Abbreviations, annotations and conventions used in the Mark Scheme	/	=	alternative and acceptable answers for the same marking point
	;	=	separates marking points
	NOT	=	answers which are not worthy of credit
	( )	=	words which are not essential to gain credit
	____	=	(underlining) key words which **must** be used to gain credit
	ecf	=	error carried forward

1.

(a)	kg m s^{-2}	B1
(b)(i)	W = 80 x 9.81 = 785 (N) (allow 784 if 9.8 is used, do not allow use of g = 10)	 B1
(ii)	horizontal force = 250 cos30 = 217 (N) (not 216 but allow 216.5)	C1 A1
(iii)	vertical component = 250sin30 = (125) force exerted on the ground = 785 – 125 = 660 (N) (allow 659 using 9.8) (allow one mark for an answer of 125)	C1 C1 A1
(c)(i)	for constant velocity resultant force is zero / in equilibrium other forces must act against / resistive / friction / opposite to the pulling force	B1 B1
(ii)	greater force exerted on the ground vertical component acts downwards (allow 2 for calculated value of 910/909)	B1 B1

2.

(a)	(i)		A newton is the <u>force</u> that causes a (mass of) one <u>kilogram</u> to <u>accelerate</u> at one m s^{-2}	B1	not just 1 kg m s^{-2}
	(ii)		Work equals the force multiplied by the distance moved <u>in the direction of the force</u>	B1	force x displacement in line of force
(b)	(i)	1	W = Fs		Allow d / x instead of s
		2	and F = ma	B1	Allow equations in words
	(ii)		W = m x a x s allow W / s = ma	B1	need W in answer line or somewhere W = KE(lost)
			a = (v-u) / t and s = [(u + v)/2] x t or a x s = $(v^2 – u^2) / 2$		
			W = m x (v-u) / t x [(v+u)/2] x t and v=0 Or W = m x $(v^2 – u^2) / 2$ and v = 0	B1	
			<u>W</u> = ½ mu^2	B1	need u here or initial velocity defined as v if v given
	(iii)		Distance is proportional to $(velocity)^2$	B1	allow double the speed causes four times distance etc
(c)	(i)		F x s = ½ mu^2 F = 0.5 x 1550 x $(13.3)^2$ / 25 = 5500 (5483) (N)	 C1 A1	 note calculating acceleration = 3.54 scores this mark allow 5480 (m) note using 1500 (-1) gives 5300 (N)
	(ii)		100 (m)	B1	

3.

(a)(i)	distance travelled per unit time / rate of change of distance	B1
(ii)	speed has magnitude only (allow size but not quantity)	B1
	velocity has magnitude and direction	B1
	(speed is a scalar and velocity is a vector scores 1)	
	(velocity has direction but speed does not scores 1)	
(b)	$t = (v - u) / a$ or $t = 2s / (u + v)$	C1
	substitution:	
	$s = (u + v)(v - u) / 2a = ((v^2 - u^2) / 2a)$	
	$v = u + 2as / (u + v)$	B1
	hence $v^2 = u^2 + 2as$	A0
(c)(i)	$v = u + at$ $u = 0$	C1
	$= (0 +) 9.81 \times 0.9$	M1
	hence $v = 8.8$ (29) m s^{-1}	A0
(ii)	if scale diagram used:	
	scale given	B1
	correct triangle drawn (shape)	B1
	velocity = 10.4 ± 0.2 m s^{-1}	B1
	angle to the horizontal = $57 \pm 2°$	B1
	if calculated:	
	triangle drawn (shape correct but arrows not required)	B1
	algebra given	C1
	velocity = 10.45 m s^{-1} (allow 10.4 or 10.5)	A1
	angle = 57.6°	A1
(iii)	vertical distance $= (8.8 \times 0.9) / 2$	C1
	$= 3.96$ m (3.97 if 8.829 is used)	A1
	(3.95 if 9.81 and 8.8 used)	
	(allow 4.0 but not 4)	
	horizontal distance $= 5.6 \times 0.9 = 5.0$ m	B1

4.

(a)	$v_x = 42 \cos 36$	**B1**
	$= 34$ (m s^{-1})	**A0**
(b)	time = 170 / 34	**B1**
	= 5 (s)	**A0**
(c)(i)	$v_y = v \sin 36 / \cos 54$	**C1**
	= 24.7 (allow 25)	**A1**
(ii)	$v^2 = u^2 + 2as$ / $s = ut + ½ at^2$ / $s = [(u + v)/2]t$	**C1**
	$0 = (24.7)^2 - 2 \times 9.81 \times s$ ecf from (c)(i)	**C1**
	s = 31.(1) (m) allow 31 to 32 as answer depends on sig figs and equation used	**A1**
(d)	k.e. = ½ m v^2	**C1**
	$= 0.5 \times 50 \times 10^{-3} \times (34)^2$	**C1**
	= 28.9 (J)	**A1**
(e)(i)	height less	**B1**
	range less	**B1**
(ii)	force acting against the motion and this effect on the acceleration / ball does work against air resistance / k.e. reduced due to air resistance / velocity is reduced by air resistance	**B1**

5.

(a)(i)	power = work done / time taken /rate of energy transfer	**B1**
(ii)	joule is the unit of work / energy used when a force of one newton moves through one metre (in the direction of the force)	**B1**
(b)(i)	potential energy = mgh / weight x height	**C1**
	= 500 x 9.81 x 3.9	**C1**
	= 19. (130) (kJ)	**A0**
(ii)	work done = 25 - 19	
	= 6 (5.870) (kJ)	**A1**
(iii)	distance up the slope = 7.8m	**B1**
	work done = force x distance	**C1**
	force = 5870 / 7.8	
	= 753 (N) (769 N if 6000 J is used)	**A1**
	note ecf from (b)(ii)	

6.

(a)(i)	speed = d / t	C1
	= 24 / 55	
	= 0.436 (m s^{-1}) allow 0.44 do not allow one sf	A1
(ii)	kinetic energy = ½ m v^2	C1
	= 0.5 x 20 x $(0.436)^2$	
	= 1.9 (J) note ecf from (a)(i)	A1
(iii)	potential energy = mg h	C1
	= 20 x 9.8 x 4	
	= 784 (J)	A1
	penalise the use of g = 10	
(b)(i)	power = energy / time or work done / time	C1
	= (15 x 784) / 55 note ecf from (a)(iii)	
	= 214 (W)	A1
(ii)	needs to supply children with kinetic energy	B1
	air resistance	B1
	friction in the bearings of the rollers / belt	B1
	total mass of children gives an average mass of greater than 20 kg	B1 Max B2

7.

)(i)	force = pressure x area / 150 x 8	C1
	= 1200 (N)	A1
(ii)	power = F x v	C1
	= 1200 x 16 ecf	C1
	= 19200 (W)	A1
(iii)	(force greater as) air resistance is greater	B1
	explanation of why: correct quantification, air resistance proportional to v or v^2 or in terms of molecules harder collisions or increased rate of collision	B1
	power greater as force is greater	B1
	power greater as velocity is greater	B1
		MAX 3

8.

$v^2 = 0 + 2 \times 9.81 \times 25$ (or using $\frac{1}{2} m v^2 = mgh$)

$v = 22(.1)\ \text{m s}^{-1}$

$22.1 = 0 + 9.81 \times t$ (or $25 = \frac{1}{2} \times 9.81 \times t^2$)

t (=22.1 / 9.81) = 2.26 s or t [=$(5.097)^{1/2}$] = 2.26 s

horizontal distance = 15 × t
= 15 × 2.257 = 33.86 (allow 15 × 2.3 = 34.5) C1

$(\text{displacement})^2 = (\text{horizontal distance})^2 + (\text{vertical distance})^2$ C1
$= (25)^2 + (33.86)^2$ C1

displacement = 42 (42.08) m (allow 43 (42.6) m, allow 2 or more s.f.) A1

distance is the actual (curved) path followed by ball
displacement is the straight line / minimum distance P to Q

9.

weight <u>vertically</u> down
normal / reaction / contact (force) perpendicular / normal <u>to the slope</u>

acceleration = gradient *or* $(v - u)/t$ *or* $\Delta v/t$
$= (6.0 - 0.8)/(2.0 - 0.0) = 2.6\ \text{m s}^{-2}$

$F = ma$
$= 65 \times 2.6$
$= 169$ N (allow to 2 or 3 s.f.)

weight component seen: $mg\sin\theta$ (218 N)
$218 - R = 169$
$R = 49$ N (require 2 s.f.)

10.

$$\text{KE} = \frac{1}{2}mv^2$$

$$= \frac{1}{2} \times 0.4 \times (30)^2$$

$$= 180\ \text{J}$$

$$s = 0 + \frac{1}{2} \times 9.81 \times (2.16)^2 \quad \textit{or}\ s = (30\sin 45°)^2/(2 \times 9.81)$$

$$= 22.88\ (22.9)\ \text{m} \qquad = 22.94\ (22.9)\ \text{m}$$

$$\text{GPE} = mgh$$
$$= 0.4 \times 9.81 \times 22.88 = 89.8\ (90)\ \text{J}$$

KE = initial KE – GPE = 180 – 90 = 90 J

(horizontal) velocity is not zero / (object) is still moving / answer explained in terms of conservation of energy

11.

loss in potential energy due to decrease in height (as P.E. = mgh)
gain in kinetic energy due to increase in speed (as K.E. = ½ mv^2)
special case 'as PE decreases KE increases' (1/2)
increase in thermal energy due to work done against air resistance
loss in P.E. equals gain in K.E. and thermal energy